Marcus O'Mara
Clojure Engineer | PhD Physicist

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Senior Software Engineer with 5+ years' experience designing and delivering distributed cloud-native systems across fintech, data engineering and edtech. Experienced across backend development, DevSecOps, cloud infrastructure and platform engineering using Clojure, Kubernetes, Kafka and AWS/GCP.

Strong track record of improving engineering productivity through automation, reducing build times by orders of magnitude, improving deployment reliability, eliminating security vulnerabilities and simplifying complex systems. PhD in Experimental Physics with a proven ability to rapidly master unfamiliar domains and solve technically challenging problems.

Passionate about building maintainable software, improving developer experience and helping engineering teams move faster through thoughtful architecture, documentation and automation.

Founder | Wiktok.io
4 Months | March 2026 - Present, Brighton

Founded and built an AI-powered educational discovery platform transforming Wikipedia into a personalised, swipeable learning experience.

- Designed and implemented a cloud-native Clojure/PostgreSQL platform supporting recommendations, AI summaries, analytics, reading lists and user management.
- Built large-scale Wikipedia ingestion and recommendation pipelines processing millions of articles with taxonomy, ranking and freshness analysis.
- Integrated Gemini and Imagen workflows with automated AI quality validation for generated summaries, artwork and voice experiences.
- Implemented production observability, Cloudflare edge infrastructure, intelligent SSR caching and comprehensive automated testing.
- Designed and delivered the platform as a scalable production system from concept through deployment.

Software Engineer | Obrizum
9 Months | June 2025 - March 2026, Cambridge

Joined following a significant engineering reduction in force, inheriting a highly distributed platform spanning over 100 repositories with limited documentation, unclear ownership and substantial technical debt. Quickly established a broad understanding of the platform, documenting architectural issues and driving engineering improvements across testing, CI/CD, infrastructure and operational processes. Deliverables include:

- Reduced end-to-end test execution from approximately **24 hours to under 30 minutes** by redesigning the integration test harness to reuse existing PostgreSQL migrations.
- Reduced AWS CDK infrastructure feedback cycles from approximately **one hour to under three minutes** through effective build caching.
- Introduced automated Trivy security scanning into CI, identifying and surfacing 100+ high and 10+ critical dependency vulnerabilities.
- Led engineering contributions for the Knowledge Space Wizard, one of two flagship strategic initiatives.
- Took technical ownership for the company's largest enterprise customer (>50,000 users), delivering additional test environments, enhanced monitoring/alerting and customer support.
- Diagnosed and fixed critical defects in the user registration flow affecting customer logins.
- Proposed and introduced structured production triage processes to reduce context switching and improve incident response across the engineering team.
- Produced extensive architectural documentation and technical recommendations, providing leadership with a clear prioritised roadmap for reducing technical debt.

Software Engineer | Gresham Technologies
2 Years 6 Months | November 2022 - May 2025, Bristol

Full-stack engineer developing a cloud-native virtual banking platform for a global fintech organisation. Alongside product development, became technical owner for application security, developer tooling and operational improvements.

Key achievements

- Reduced deployment time by **50%** by identifying thread-safety issues and parallelising integration test execution.
- Improved CI stability by approximately **20%** through distributed tracing and optimisation of database queries.
- Eliminated all Critical and High container vulnerabilities across production deployments.
- Designed and built internal tooling to visualise complex Kafka Streams topologies, dramatically improving developer onboarding.
- Standardised production incident investigations through improved documentation and knowledge sharing.
- Developed monitoring, automation tooling and operational dashboards to improve supportability and developer productivity.

Tech-stack: Clojure, Babashka, GraphQL, Kafka, PostgreSQL, Docker, Kubernetes, Helm, Kustomize, Terraform, GCP, CircleCI, Grafana, Prometheus, Keycloak.

Data Engineer | The Hyde Group
9 Months | September 2021 - November 2022, London Bridge

Designed and delivered a cloud-native document intelligence platform processing approximately one million physical documents using AWS machine learning and data engineering services.

Key achievements

- Architected an automated OCR and NLP pipeline using AWS Textract, Comprehend and SageMaker.
- Enabled extraction of previously inaccessible business data to improve reporting and operational efficiency.
- Delivered event-driven integrations between Salesforce, Workday and internal systems using Kafka and AWS.
- Contributed to the organisation's migration from legacy platforms towards cloud-native microservices.

Tech-stack: AWS (Lambda, EC2, ECS, S3, Redshift, Athena, CloudWatch, Textract, Comprehend, SageMaker), Kafka, Docker, Terraform.

Software Engineer | Allstreet
10 Months | November 2020 - September 2021, Brighton

Developed features for a VC-backed fintech platform specialising in financial sentiment analysis and market intelligence.

Key achievements

- Built frontend and backend functionality using Clojure and ClojureScript.
- Developed high-volume web scrapers covering major stock exchanges and the SEC EDGAR database.
- Extended REST APIs supporting interactive financial visualisations.
- Increased automated test coverage through extensive integration and UI testing.
- Reduced technical debt and mentored junior developers.

Tech-stack: Clojure, ClojureScript, React, Re-frame, MongoDB, Elasticsearch, Kafka, D3, Material UI, AWS.

Doctor of Philosophy | University of Sussex
3 years 9 Months | September 2017 - June 2021, Brighton

PhD in Experimental Nanotechnology | Roger Blin-Stoyle Award for Outstanding Thesis

Thesis: “Functional Composites from Graphene Stabilised Silicone Emulsions”

Designed, engineered, tested and modelled advanced materials for health-care, soft robotics and sensing applications. The patent pending technology developed redefined the state-of-the-art, and received additional funding to provide additional telemetry data on parachute deployment status in partnership with NASA.

Publications: Ultra-sensitive strain gauges enabled by graphene-silicone emulsions | Nanosheet-stabilised emulsions: ultra-low loading segregated networks and surface energy determination of pristine few-layer 2D materials | Large-scale surfactant exfoliation of graphene and Conductivity-Optimised graphite enabling wireless connectivity | Graphene-based printable conductors for cyclable strain sensors on elastomeric substrates | Size selection of liquid-exfoliated 2D nanosheets

Associate Tutor | University of Sussex

10 Months | January 2020 - June 2020 | September 2018 - December 2018 Brighton

Led workshops aimed at undergraduate students for the *Introduction to Nanomaterials and Characterisation* and *Foundation Mathematics* modules. Key competencies include problem solving, active listening, clear communication and leadership.

Research Support Technician | Advanced Materials Development

10 Months | February 2019 - December 2019, Brighton

Designed and implemented processes for the production of high-quality nanomaterial dispersions. This involved understanding and solving production bottlenecks through process optimisation, assessing impact on product quality using advanced analytical techniques and associated cost-benefit analysis.

Web Designer | Freelance

4 Years | 2018 - 2022

Creating websites for small companies in the south east. Responsibilities include business generation through brand development and targeted advertising.

Theoretical Physics, MPHYS, 1st Class Honours | University of Sussex

4 Years | September 2013 - July 2017, Brighton

Dissertation: Loss and Phase Estimation Using Entangled and Unentangled States

Constructed a Bayesian statistical simulation in MATLAB to investigate the possibility of entangling photons to enhance measurement precision of quantities of interest. This work extended research I undertook over summer 2016, culminating in a Sussex Undergraduate Research Award.

Key Modules: Scientific Computing, Programming in C++, Mathematical Methods I, II, III, Quantum Mechanics I, II, Further Quantum Mechanics, Analysis I, II, Complex Analysis, Thermal and Statistical Physics, General Relativity, Quantum Field Theory, Introduction to Nano-materials and Nano-characterisation, Theoretical Physics, Atomic Physics, Nuclear and Particle Physics, Particle Physics, Condensed State Physics

Simon Langton Grammar School | 3 A-levels | 2 AS-levels

September 2011 - July 2013, Canterbury

B B B A C - Maths, Physics, Economics, French, Further Maths

Simon Langton Grammar School | 10 GCSEs

September 2006 - July 2011, Canterbury

3 A*'s, 6 A's, 1B - English Literature, Geography, Chemistry, Physics, Maths, Economics, English Language, French, Biology, ICT